Comparison of Perceived Brightness and Colorfulness for Different (Automotive) Display Technologies by Helmholtz-Kohlrausch Effect

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Abstract
Modern cars feature different display technologies with specific color coordinates. The Helmholtz–Kohlrausch (H-K) effect describes that more saturated colors appear brighter for the same luminance. We evaluated three LCDs (2 standard and 1 QD backlight) and one RGB OLED by 20 subjects. The H-K effect was verified. Other results included 8% luminance sensitivity for tiled displays, the need for color rendering for photos, and that the RGB OLED provides the deepest black in direct comparison.

Author Keywords
Helmholtz-Kohlrausch; brightness; luminance; gamut; saturation; LCD; OLED; Quantum Dot

1. Introduction
Displays are widespread especially in modern cars, are increasing in size (including pillar-to-pillar, Figure 1) and are growing in annual volume[1]. Furthermore, LCDs (edge-lit, local dimming and quantum dot backlights) are competing with OLEDs, and will do so with MicroLEDs in the future. Different display technologies typically have different color coordinates (saturation, purity), resulting in varying perceived brightness, readability, and colorfulness. This phenomenon is explained by the Helmholtz-Kohlrausch (H-K) effect (see references[2 to 6]) and visualized in Figure 2. Even if the luminance remains the same, colors appear brighter when purity increases (e.g. when they are closer to the horseshoe-like curve of CIE 1931 for monochromatic light), and they appear more colorful[4]. This is essential for the calibration of modern cockpits with various displays and light sources.

We examined this phenomenon using a specialized setup to compare automotive display technologies. Our goal is to provide additional methods for selecting and calibrating displays.

Figure 1. Premium automotive cockpit with LCD (instrument cluster), OLED (center and codriver display) and LED-based interior lighting. Source: Mercedes.

2. Displays and Setup for H-K Evaluation
We compared four displays with different color coordinates that are currently used in cars:

(1) TN LCD for low budget vehicles
(2) IPS LCD with wide viewing angle (mostly used 2025)
(3) IPS LCD with Quantum Dot backlight film (demos)
(4) RGB OLED in mass production for premium vehicles

The display types were not known to the subjects. These samples have an aspect ratio of 16:9 and sizes ranging from 10” to 11”. They were adapted to 10” using a checkerboard mask. The maximum common color gamut (CCG, see Figure 3) is the gamut that can be rendered on all four displays with their specific measured color coordinates. Test patterns (see Figure 4) were calibrated to this CCG with D65 as white point including a gamma (γ) value of 2.2 (relevant for GUIs and photos). Manual 'fine tuning' was performed to achieve the same luminance, for example, by modifying the RGB grey levels via lightness using the HSL model.

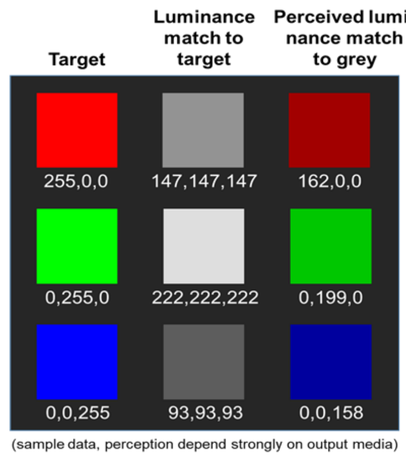

Figure 2. Visualization of the Helmholtz-Kohlrausch (H-K) effect (adapted[6]): (Measured) luminance (center and left) vs. perceived brightness (human vision, right) for red, green and blue. The numbers refer to 8-bit grey scale.

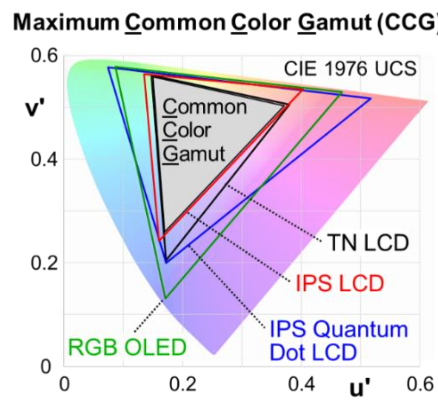

Figure 3. Color gamut of the four displays under test and their (maximum) common color gamut (CCG, covered by “all” displays) for CIE 1976 UCS.

Figure 4 shows the setup for the evaluation: The four displays were mounted behind a vertical fixture, leaving a 40 mm distance due to the bezels. A random monochrome checkerboard surrounds them to avoid potential human eye disturbance. We used an illumination of 150 lux for the display wall in a dark room without reflections on the displays (L_{White} = 80 cd/m² for CCG) as

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well as dark room conditions (< 1 lux). The observer distance was set to 1 m, with the eye height adjusted to the center of the displays (gaze point) to minimize potential viewing angle effects.

Four identical PCs, each with the same graphics adapter, drive the four displays. We used three different categories of test patterns, which are visualized in Figure 4. During the evaluation, the same test patterns were shown in parallel on the four displays:

- Full-screen test patterns for RGBW and CMY.
- Automotive GUI with typical icons and colors using CCG. The colors of the full-screen test patterns were used to create the icons. GUIs of the instrument cluster and maps did not produce useful results in our pre-test.
- Photos: Two female faces and foliage (colored leaves).

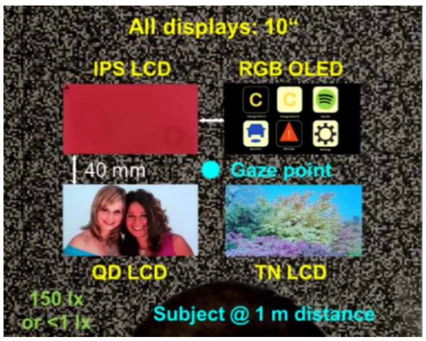

Figure 4. Set-up for evaluation with four displays for the Helmholtz-Kohlrausch effect. The displays show here the different test patterns used in our test (same image during evaluation). The subjects did not know the display types.

Different methods were used to create the test patterns; examples are illustrated in Figure 5 for full-screen tests:

(1) Luminance calibration for CCG: The same measured luminance was used for CCG full-screen test patterns and the automotive GUI; both are shown at the top of Figure 4.
(2) Brightness calibration for CCG: Adjusted to the same perceived brightness by three experts for CCG for full-screen test patterns and automotive GUI. Luminance (and color values) were measured and noted.
(3) Luminance calibration for maximum saturation: Maximum saturated RGB (one grey level to adjust luminance, the other grey levels for the primaries are zero) for full-screen.
(4) Luminance calibration for photos: Done for photos using CGG and maximum saturation (purity).
(5) Luminance calibration for photos: Various renderings of the photos with differently reduced saturation and different white points, for comparison of the QD LCD and RGB OLED with the IPS LCD (judged as 'natural' in a pre-test).

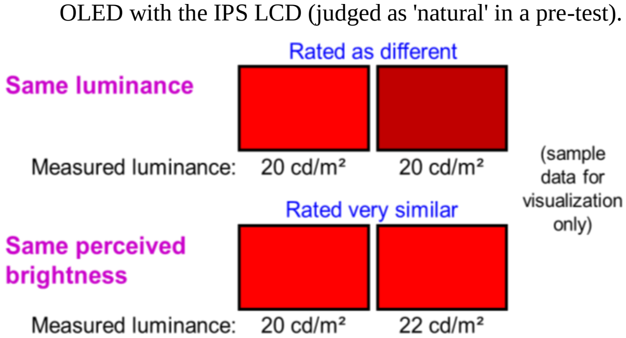

Figure 5. Visualization as example of the full-screen test pattern types for luminance and brightness calibration.

3. Evaluation Results

We conducted a user study with 20 participants after refining the test patterns according to a pre-test, which also involved 20 participants, 50% of those were different from the final test. 65 % were between 20 and 30 years old (the rest were older) and 80 % were male (we do not expect gender to have an impact). 35 % were experienced in display evaluation. 80 % of the subjects were judged by the observer as to be relaxed, focused and credible in their responses (in this combination). The rest had at least one of these three criteria that was less favorable, but not too much. The display types were not known to the subjects. The evaluation method was in accordance with ITU-R BT.500-13: The double-stimulus impairment scale (DSIS) method involved five steps for the direct comparison of neighboring displays as well as the ranking of the displays from brightest to darkest (all using a forced-choice approach). The tasks of the subjects were:

A. Rank all four displays from brightest to darkest for full screen test patterns and judge on vividness for full saturation.
B. Compare directly (full screen patterns) the QD LCD and the RGB OLED with the IPS LCD regarding which one is brighter and how large the difference is perceived.
C. Automotive GUI: Compare the readability of the icons, judge on vividness of the colors, which gear wheel of the icon ‘settings’ is the darkest …
D. Photos: Judge on authenticity for maximum saturation and select for the QD LCD and the RGB OLED the sample of various renderings, which is closest to the IPS LCD.

The following sections present selected results.

3.1 Luminance-calibrated CCG full screen - method ‘A’

Blue was ranked as brightest in Figure 6 for the OLED (highest saturation see Figure 3) by 45% of the subjects by method ‘A’ (see above). This is in accordance with the color coordinates (purity) shown in Figure 3 for the four displays and confirms the Helmholtz-Kohlrausch effect. The differences observed for green and white were smaller than those for blue.

Figure 7 shows the ranking results for red using ‘A’ and a weighting function: Sum of the ranks (brightest ‘1’ to darkest ‘4’) weighted by 8 for #1, by 4 for #2, by 2 for #3 and by 1 for #4. The standard deviation of the ranking for saturations over 90% was approximately ±10 for a weight of >60. The results are not fully compliant with the H-K effect as the IPS LCD was rated as brightest but the OLED has a higher saturation.

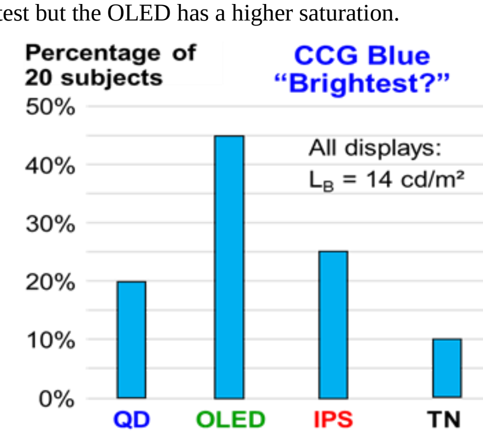

Figure 6. Selected evaluation results: Common color gamut (CCG) blue using luminance calibration as reported by the subjects.

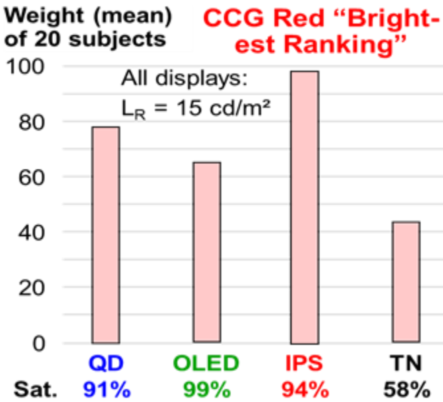

Figure 7. Selected evaluation results: Red using common color gamut (CCG) for luminance calibration with weighted ranking and saturation.

3.2 Luminance-calibrated CCG full screen - method '<u>B</u>'

In addition, we asked the subjects to make direct comparisons of the neighboring displays, asking "Which one is brighter?" and "How large is the perceived difference?" (method 'B', see above). The conditions were CCG and luminance-calibration. The TN LCD was excluded as a result of the pre-tests. Figure 8 left visualizes the direct comparisons for blue between the IPS LCD and the QD LCD: The majority rated the perceived brightness as the same or only very slightly different. 4 of 20 subjects stated that the IPS is (as mean) very slightly to slightly brighter than the QD LCD. Only 10% reported the QD to be brighter than the IPS. Figure 8 right shows the corresponding results for IPS LCD vs. the OLED: 45% ranked the OLED to be brighter (very slightly to slightly) and only a few vice versa. One third perceived both displays as being the same. Those results correspond with the "forced" ranking shown in Figure 6, thereby validating both methods and the Helmholtz-Kohlrausch effect for this case.

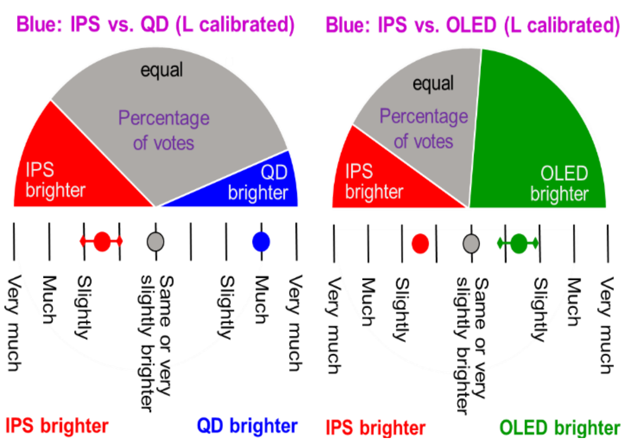

Figure 8. Direct comparison of IPS LCD to QD LCD (left) and OLED (right) for CCG blue and luminance calibration.

3.3 Perceived brightness -calibrated CCG full screen

As expected, the brightness-calibrated test patterns for CCG show fewer variations in ranking and smaller perceived differences in direct comparison for neighboring displays than the luminance-calibrated patterns. The outcome of the brightness-adjusted tests enables the measured luminance differences to be judged. Table 1 shows the measured luminance values for the primaries incl. some modifications ('modified') to force perceived differences (outcome of the pre-test). The TN LCD is omitted here as being of low relevance for modern cars.. The subjects were asked to provide rankings (method 'A') and to perform direct comparisons (method 'B'), which is the focus here. The IPS LCD at 21.8 cd/m² for 'red modified' is perceived as roughly the same brightness as the QD display at 23.1 cd/m². When the luminance for 'red' of the IPS LCD is reduced to 19 cd/m² (a difference of ~4 cd/m²) the QD LCD is clearly perceived as being brighter. Consequently, a difference of ~ 3 cd/m² at a distance of 40 mm between the displays was clearly noticeable to the participants. This helps to define a threshold for the luminance calibration of tiled displays. Green leads to less significant results as expected by the H-K effect. The OLED appeared slightly brighter overall for ~13 cd/m² for 'blue modified'. A 'rule of thumb' from these tests are that 40 mm tiled displays should differ by less than 10% in luminance to achieve the same perceived brightness.

Table 1. Measured luminance values for brightness calibrated test patterns incl. variations (modified: highlighted in purple) to force differences.

L /cd/m²	IPS LCD	QD LCD	OLED
Red	19.0	23.1	19.1
Red modified	21.8	23.1	19.1
Green	63.7	68.0	59.1
Blue	13.0	13.9	15.0
Blue modified	13.0	13.9	13.2

3.4 Automotive GUI at various settings

There is hardly any observed difference for the automotive GUI (see top right of Figure 4) between luminance and brightness calibration for CCG rendering. This is due to the icons being much smaller in size than the full-screen test patterns, as well as being spaced further apart. A preference for e.g. highly saturated red for warnings was stated. This has also advantages at bright light conditions ("bleaching of colors", see reference[4] as the mixed color (original plus white light) has a higher saturation. The perception of a gear wheel (surrounded by the same white luminance for all displays) of the settings icon of the automotive GUI (see Figure 4 top right): 80% responded that the OLED showed the darkest black (Figure 9), while 95% judged the TN LCD to have the brightest black (not plotted here). This clearly proves that the 'true black' of OLEDs is perceived as such even when surrounded by white (the local contrast is highest).

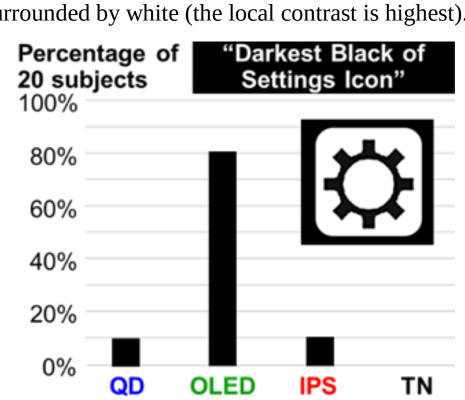

Figure 9. Automotive GUI: Perception of black of the settings icon is judged as best (deepest black) for the OLED.

3.5 Additional Findings

- There was no significant difference in the ranking for L-calibrated CCG for the RGB for full screen test patterns between 150 lux and dark room conditions.
- There was no significant difference in brightness perception for full screen white for both CCG and maximum saturation. The differences for green for these cases were much smaller than for red and blue. This is in accordance with the H-K effect.
- Photos (see bottom of Figure 4) should be rendered to smaller gamut for large gamut displays (QD LCD, RGB OLED). They were judged at full saturation as unnatural, see Figure 10. Very similar judging was provided for the image showing leaves (foliage). The TN LCD had noticeable degradation for photos due to gamma issues.

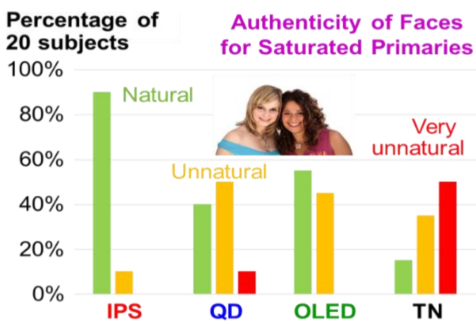

Figure 10. Authenticity of photos (faces) for saturated colors and luminance calibration.

4. CIECAM Simulation of Helmholtz-Kohlrausch

We investigated as well for the displays under test and the environment the CIECAM16 model[7] as improved by Hellwig[8] to include H-K effect on perceived brightness. The adapting luminance is 5 cd/m². This simulator can apply many conditions that can consider surround luminance and display white point that will change as the viewing environment changes. In all cases, two fundamental truths emerge see Figure 11: The QHK (H-K-modified brightness) exhibits a stronger effect than the traditional brightness (Q, grey bars) for the primaries. In addition, the technologies with the widest color gamut, such as the OLED or Quantum Dot LCD as shown in Figure 3, produce the strongest H-K effect. However, the difference for white (right) between Q and QHK is very small compared to the primaries RGB. This is an expected and validated result.

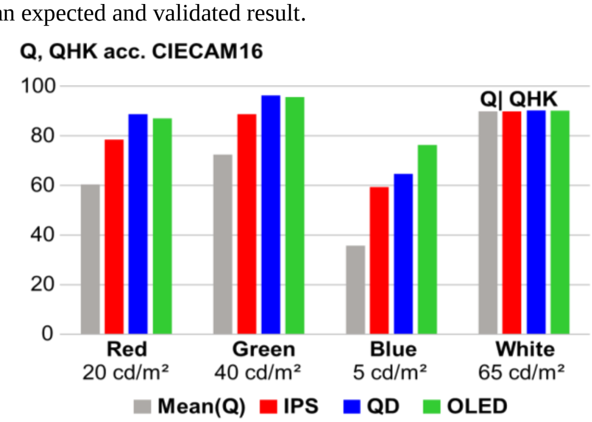

Figure 11. Simulation results of brightness Q vs. H-K compensated brightness QHK acc. CIECAM16[7] for luminance calibration for three displays.

Our evaluations confirm the Helmholtz-Kohlrausch effect (except for some deviation for red), whereby colors with the same CIE coordinates appear brighter when generated by higher saturation primaries. For the QD LCD, red has the highest simulated brightness QHK, compare to Figure 3. However, the evaluation shows that the IPS LCD is ranked as the brightest for the same luminance and CCG, see Figure 7 and the other tests (5% luminance difference, see §3.3), which are not shown in more detail here. We will further study this to determine whether the CIECAM Helnholtz-Kohlrausch model should be improved or if this deviation is within acceptable limits for evaluations involving subjects.

5. Summary

We have successfully implemented a test setup that compares display technologies in terms of the perceived brightness and colorfulness of various colors, automotive GUIs, and photos. Our goal was to evaluate and to judge on the Helmholtz-Kohlrausch effect, which we achieved (with additional results):

Displays with a larger gamut (higher saturation) appear brighter at the same (measured) luminance. This can be used to reduce power consumption and to improve color perception in the presence of ambient light reflections.

Furthermore, photos should be rendered on large gamut displays to a smaller gamut (like standard LCDs). Tiled displays at 40 mm distance should differ in luminance by less than 10%.

Our findings will help to design, to calibrate and to improve displays in addition to “traditional” parameters such as luminance and cost for use in various applications such as vehicles.

6. Acknowledgements

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7. References

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